A two-level oral-biofilm model from one variational principle

derivation, discretisation, verification and first results

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Abstract

In the thesis the growth field of Klempt et al. 2024 and the multispecies point model of Klempt et al. 2026 are solved separately and joined at the Gauss points, which needs choices that no paper fixes: how the two voids relate, whether the point model is scaled to the field, what drives the growth, and how the time of the point model is linked to the time of the field. Here the biofilm is described on two levels, a phase field ϕ (biofilm against the outer liquid) and, inside the biofilm, the composition of the point model, and all equations follow from one energy, one dissipation potential and one constraint through the extended Hamilton principle in Onsager's form. Growth is a reaction that consumes nutrient, catalysed by the living biomass. The model recovers the point model where the field is uniform, the element's scheme (c) as its decoupled limit, and the motion of the biofilm edge towards the nutrient without the front term of Klempt et al. 2024. The total energy decreases along solutions. Every discrete equation solved is the automatic derivative of the discrete functional, and the implementation is checked against the point model to round-off. In 3D with the calibrated five-species point model, an antibiotic lowers the living biomass, the growth and the advance of the edge, differently for a broad and an anaerobe-specific antibiotic, and the composition varies with the local nutrient.

In plain words

The problem. In the thesis two models are computed separately and joined afterwards: the field says where the biofilm is and how much of it there is, and the point model says which species are inside and what fraction of each is alive. Joining them needs choices that no paper makes: whether to scale the point model to the field (schemes (a) and (b)), and what makes the biofilm grow (schemes (a) and (c)).

What is done here. Both are written as one model from the start, and every equation follows from one principle: the system moves so that its energy goes down, at a rate set by friction (dissipation). Nothing is joined afterwards, so the choices above disappear.

What comes out.

1. *There are two kinds of empty space.* The liquid outside the biofilm and the liquid inside it are different things. A model that mixes them lets bacteria grow by themselves in the outer liquid.
2. *Growth is a reaction that eats nutrient.* Living cells run it, so when an antibiotic kills cells, the growth slows down. Writing the living cells as the driver of the rate, not as an energy, is what makes the antibiotic work in the model.
3. *The biofilm spreads towards the nutrient by itself.* Where there is more nutrient the biofilm grows more and pushes its edge out further. Klempt et al. 2024 added a separate term for this; here it follows from the growth, in 1D, 2D and 3D.

In one sentence. The thesis's scheme (c), growth from the living biomass, is the simplest limit of this model, so it now has a derivation behind it.

How to read the rest. Sections 1–5 give the model and its derivation, Section 6 the numerics, Section 7 the checks, Section 8 the results, Section 9 what is assumed and open.

1 Why two levels

The direct generalisation of the point model to space takes every species as a field $\phi_i(\mathbf{X},t)$ with one void $\phi_0 = 1 - \sum_i \phi_i$ and the point-model energy at every point. The point model keeps every ϕ_i away from zero by its barrier energy $K/[\phi_i^2(1-\phi_i)^2]$, and its production is local. Solved in 1D with all fields in one Newton system (`unified_1d_monolithic.py`), the outer liquid fills up from the barrier floor everywhere at once (at $T^* = 1$ the amount far from the seed had risen from 0.05 to 0.43), so the model cannot represent a biofilm next to liquid. The two voids are therefore different physical objects: the liquid outside the biofilm, which the field of Klempt et al. 2024 describes, and the liquid inside it, which the point model's ϕ_0 describes. The model below keeps both.

2 State

symbol	meaning	source
$\phi(\mathbf{X},t) \in [0,1]$	biofilm phase: biofilm against the outer liquid	Klempt et al. 2024
$s_i(\mathbf{X},t)$, $i = 1 \dots n$	share of species i in the biofilm	Klempt et al. 2026 (ϕ_i there)
$s_0(\mathbf{X},t)$	inner liquid, $\sum_i s_i + s_0 = 1$	Klempt et al. 2026 (ϕ_0 there)
$\psi_i(\mathbf{X},t)$	living fraction of species i	Klempt et al. 2026
$\gamma(\mathbf{X},t)$	multiplier of the composition constraint	Klempt et al. 2026
$c(\mathbf{X},t)$	nutrient concentration	Klempt et al. 2024
$\alpha(\mathbf{X},t)$	local expansion, $\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$	Klempt et al. 2024
$\mathbf{u}(\mathbf{X},t)$	displacement	Klempt et al. 2024

The absolute amount of species i is ϕs_i and the living biomass is $\phi \sum_i s_i \psi_i$. The point model's $\bar{\phi}_i = \phi_i \psi_i$ is $s_i \psi_i$ here. All fields live in the reference configuration $\mathbf{X}$.

3 Energy, dissipation, constraint

Let $h(\phi) = \phi^2$ be the weight with which the biofilm carries its composition, and $w = h(\phi) + \varepsilon$ with a small $\varepsilon > 0$ that keeps the composition equations regular in the outer liquid, where they carry no weight.

Composition. The energy of the point model of Klempt et al. 2026, per unit biofilm,

$$\Psi_{\mathrm{pm}} = \underbrace{K\Big[\sum_i B(s_i) + B(s_0) + \sum_i B(\psi_i)\Big]}_{\Psi_{\mathrm{bar}}} + \underbrace{\frac{\alpha^*}{2}\sum_i b_i \psi_i^2 - \frac{c^*}{2}\sum_{ij} A_{ij}\, s_i \psi_i\, s_j \psi_j}_{\Psi_{\mathrm{int}}}, \qquad B(x) = \frac{1}{x^2(1-x)^2}, \tag{1}$$

with the barrier constant $K = 10^{-4}$, the interaction matrix A (calibrated, symmetric), the nutrient constant c^*, the antibiotic concentration α^* and the susceptibilities b_i. The barrier terms keep s_i, s_0, ψ_i inside $(0,1)$. Both parts are stored energy and carry the weight w (Remark 1).

Phase, nutrient, growth. With the gradient energies of Klempt et al. 2024, a one-sided barrier that keeps $\phi \le 1$, the free energy of an ideal solute for the nutrient and the chemical potential μ_b of grown biomass,

$$E = \frac{\beta}{2}|\nabla\phi|^2 + \frac{K_\phi}{(1-\phi)^2} + \lambda w \Psi_{\mathrm{pm}} + w\frac{\beta_s}{2}\sum_i |\nabla s_i|^2 + c(\ln c - 1) + \mu_b \xi + E_{\mathrm{mech}}, \qquad \xi = \alpha - 1. \tag{2}$$

λ is the energy scale of the composition relative to the phase, β_s the spreading of the composition inside the biofilm, and $E_{\mathrm{mech}} = \frac{\mu}{2}\phi^2(\mathbf{I} : \mathbf{C}_e - 3)$ the mechanical energy of Klempt et al. 2024, which does not enter the growth equations (as there: its effect on growth is small).

Dissipation. Quadratic in the rates, from Klempt et al. 2026 for the composition and Klempt et al. 2024 for the phase, and for the two fluxes of the growth reaction:

$$\mathrm{D} = \frac{\eta_\phi}{2}\dot\phi^2 + \lambda w \mathrm{D}_{\mathrm{pm}} + \frac{\dot\xi^2}{2L} + \frac{|\mathbf{j}|^2}{2Dc}, \qquad \mathrm{D}_{\mathrm{pm}} = \sum_i \Big[\frac{\eta}{2}\dot s_i^2 + \frac{\eta}{2}\big(\tfrac{\mathrm{d}}{\mathrm{d}t}(s_i\psi_i)\big)^2\Big] + \frac{1}{2}\dot s_0^2, \tag{3}$$

where $\mathbf{j}$ is the nutrient flux and L the mobility of the growth reaction. The living biomass is its catalyst,

$$L = L_0\,\phi\sum_i s_i\psi_i, \tag{4}$$

so lysis or an antibiotic lowers the growth through the mobility and exerts no force on ψ_i (Remark 2).

Constraints. The composition constraint with the multiplier field, $\lambda w\gamma(\sum_i s_i + s_0 - 1)$, and the balance of the growth reaction, which turns Y units of nutrient into one unit of growth,

$$\dot c + Y\dot\xi + \nabla\cdot\mathbf{j} = 0. \tag{5}$$

4 The principle and the evolution equations

For the state $q = \{\phi, s_i, s_0, \psi_i, \gamma\}$ the extended Hamilton principle of Junker and Balzani (as used by Klempt et al. 2024) states

$$\delta_q \int_\Omega E\,\mathrm{d}V + \int_\Omega \frac{\partial \mathrm{D}}{\partial \dot q}\,\delta q\,\mathrm{d}V = 0 \qquad \forall\,\delta q, \tag{6}$$

with $\mathbf{n}\cdot\nabla\phi = \mathbf{n}\cdot\nabla s_i = 0$ on the boundary; the dissipation is varied with respect to the rates only. For the reaction and the flux it is Onsager's form: $\dot\xi$ and $\mathbf{j}$ minimise the Rayleighian $\frac{\mathrm{d}}{\mathrm{d}t}\int E + \int \mathrm{D}$ under the balance (5).

Composition (each equation divided by $\lambda w > 0$):

$$KB'(s_i) + \gamma + \eta(1+\psi_i^2)\dot s_i + \eta s_i\psi_i\dot\psi_i - c^*\psi_i\sum_j A_{ij}s_j\psi_j - \frac{\beta_s}{\lambda w}\nabla\cdot(w\nabla s_i) = 0, \tag{7}$$

$$KB'(s_0) + \gamma + \dot s_0 = 0, \tag{8}$$

$$KB'(\psi_i) + \alpha^* b_i\psi_i + \eta s_i\psi_i\dot s_i + \eta s_i^2\dot\psi_i - c^* s_i\sum_j A_{ij}s_j\psi_j = 0, \tag{9}$$

$$\textstyle\sum_i s_i + s_0 - 1 = 0. \tag{10}$$

Without the gradient term these are the point model of Klempt et al. 2026 at every point, in its own time.

Phase:

$$\eta_\phi\dot\phi - \beta\nabla^2\phi + \frac{2K_\phi}{(1-\phi)^3} + \lambda h'(\phi)\Psi_{\mathrm{pm}} + h'(\phi)\frac{\beta_s}{2}\sum_i|\nabla s_i|^2 = 0. \tag{11}$$

The composition acts on the phase through $\lambda h'(\phi)\Psi_{\mathrm{pm}}$: where the composition energy is low (strong favourable interactions) the biofilm tends to grow, where it is high (an antibiotic, $\alpha^* b_i\psi_i^2/2$) it tends to recede. Because $h'(0) = 0$, the composition exerts no force where there is no biofilm.

Growth and nutrient. With $\mu_c = \partial_c E = \ln c$, the Rayleighian under (5) gives

$$\dot\xi = L(Y\ln c - \mu_b), \qquad \mathbf{j} = -D\nabla c, \qquad \dot c = D\nabla^2 c - Y\dot\xi, \tag{12}$$

with $c = 1$ on the supplying boundary and $\mathbf{n}\cdot\nabla c = 0$ elsewhere. Writing $\mu_b = Y\ln c_{\mathrm{stop}}$, growth stops where c falls to c_{stop}. The kinematics is that of Klempt et al. 2024, $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$, $\mathbf{F}_g = \alpha\mathbf{I}$.

Mechanics. The principle of virtual work with the stiffness $(\phi^2+f)E_{\text{bio}}$ of Klempt et al. 2024, unchanged.

Remark 1 (the barrier terms)**.** *In the one-level point model only the derivatives of* Ψ_{bar} *matter. Weighted with the biofilm amount, its value also acts on the phase,* $\lambda h'(\phi)\Psi_{\text{bar}}$*. At the calibration's start (*$\psi_i = 0.999$*) this value is about* 500*, against about* 8 *for the interactions; after the initial transient (*$\psi_i \approx 0.9$*) it is below* 0.1*. Two alternatives were tried, holding the barrier weight fixed (a regularised constraint) and giving it a constant weight; both change* α *by about* 6 % *and the first breaks the exact energy inequality. The literal form, barrier terms as stored energy with the weight w, is kept: it is what Klempt et al. 2026 wrote, and it makes Proposition 1 exact.*

Remark 2 (growth energy or growth reaction)**.** *A first version drove the growth by an energy* $-k_\alpha(\alpha-1)\phi\sum_i s_i\psi_i$*, which gives* $\dot\alpha = k_\alpha\phi\sum_i s_i\psi_i$ *directly. Its derivative with respect to* ψ_i *is a force that keeps cells alive where growth has happened; at large growth it defeats an antibiotic (at* $k_\alpha = 8$ *the antibiotic lowered the growth by* 0.7 % *only). The reaction form* (12)–(4) *puts the living biomass into the mobility, where it belongs: a catalyst changes the rate, not the energy. It also replaces* k_α *by quantities with a meaning (*L_0*,* Y*,* c_{stop}*) and makes the energy inequality include the nutrient.*

5 Properties

Proposition 1 (energy inequality)**.** *Let* E *not depend on time explicitly and let the boundary be closed (*$\mathbf{n}\cdot\mathbf{j}=0$*). Along solutions of* (7)–(12)*,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_\Omega E\,\mathrm{d}V = -\int_\Omega \Big[2\lambda w \mathrm{D}_{\text{pm}} + \eta_\phi\dot\phi^2 + \frac{\dot\xi^2}{L} + \frac{|\mathbf{j}|^2}{Dc}\Big]\,\mathrm{d}V \le 0.$$

Proof. Multiply each stationarity condition of (6) by its rate, integrate and add. The gradient terms integrate by parts to the rates of the gradient energies (the boundary terms vanish by the no-flux conditions), the constraint terms give $\lambda w\gamma\frac{\mathrm{d}}{\mathrm{d}t}(\sum_i s_i + s_0) = 0$, and D_{pm} and $\dot\phi^2$ are quadratic in the rates, so the rates times the dissipative forces give twice the dissipation. For the reaction, $\partial_c E\,\dot c + \mu_b\dot\xi = \mu_c(-Y\dot\xi - \nabla\cdot\mathbf{j}) + \mu_b\dot\xi = -(Y\mu_c-\mu_b)\dot\xi + \mathbf{j}\cdot\nabla\mu_c$ after integration by parts, and (12) gives $Y\mu_c-\mu_b = \dot\xi/L$ and $\nabla\mu_c = -\mathbf{j}/(Dc)$. □

Remark 3. *A supplying boundary (*$c=1$*) and a switch of the antibiotic (*α^* *changing in time) are external inputs: the first adds the inflow of nutrient energy, the second* $\int\lambda w\frac{1}{2}\dot\alpha^*\sum_i b_i\psi_i^2$*. Both appear in the computations as the only increases of the discrete energy.*

Limits.

1. Uniform field, $\beta_s = 0$: at every point the composition is the point model of Klempt et al. 2026 in its own time. The calibrated A, c^* and the time link of the calibration carry over unchanged.
2. $\lambda\to 0$, $\beta_s = 0$: the composition decouples from the phase and runs where the biofilm is, as the element's free point model (scheme (b)); the growth $\dot\alpha = L_0\phi\sum_i s_i\psi_i(Y\ln c-\mu_b)$ is the element's scheme (c), $\dot\alpha = \kappa\phi_{3D}\sum_i\bar\phi_i$, with $\kappa = L_0(Y\ln c-\mu_b)$. Scheme (c) is thus the decoupled limit of this model, and its point model, which runs unscaled, is the consistent one: its ϕ_0 is the inner liquid s_0.
3. Scheme (a), which scales the point model to $\sum_i\phi_i = \phi_{3D}$, identifies the inner liquid with the outer one; it is not a limit of the model.
4. One species, $\psi\equiv 1$, constant c: $\dot\alpha = L_0\phi\,(Y\ln c-\mu_b)$ is Eq. 36 of Klempt et al. 2024 with $k_\alpha = L_0(Y\ln c-\mu_b)$.

Proposition 2 (the front moves by the kinematics)**.** *Let a biofilm layer on a substrate expand only normal to it. Its edge in the current configuration is at* $x(t) = \int_0^{X_e}\alpha^3(X',t)\,\mathrm{d}X'$*, where* X_e *is the edge in the reference configuration; the edge advances at the rate of the integrated growth, as in Wanner and Gujer's model, where the front speed is the integral of the production.*

Proof. $\mathbf{F}_g = \alpha\mathbf{I}$ gives the volume ratio $J=\alpha^3$; with expansion only in the normal direction the stretch in that direction is J, so $\mathrm{d}x = \alpha^3\,\mathrm{d}X$. □

Klempt et al. 2024 add a front term h^*_ϕ to the phase equation, which is not derived from a potential (their Remark 1). In this model the edge moves by the growth itself; adding the front term would count the same growth twice. In 2D and 3D the direction follows from the nutrient: where c is higher the growth is larger and the edge advances more (Section 8).

6 Discretisation and solution

One spatial dimension (nodes with mirrored ends for the no-flux conditions, nodal quadrature), implicit Euler in time. For one step from U^n to U the discrete functional

$$\Pi(U) = E_h(U) + \Delta t\, \mathrm{D}_h\Big(\frac{U-U^n}{\Delta t}; U\Big) + C_h(U)$$

is written down (energy, dissipation with its weights and coefficients at the new state held fixed, constraint), and the residual is its automatic derivative (JAX), $R(U) = \partial_U \Pi$, with the rows scaled to the form of the point model. The discrete equations are therefore the discrete Euler–Lagrange equations by construction; no equation is typed by hand (`unified_two_level_1d.py`, `unified_two_level_fuel_1d.py`). The Jacobian is the automatic derivative of R (including the dependence through the frozen weights). Newton's method with a damping that keeps $s, \psi \in (0,1)$, $0 < \phi < 1$ and $c > 0$, to $|R|_\infty < 10^{-8}$; a step that does not converge is halved recursively.

The 2D and 3D checks of the front (`fuel_front_2d.py`, `fuel_front_3d.py`, `fuel_front_3d_species.py`) use the decoupled limit: the nutrient and the growth (12) on a regular grid with the point model at every node of the biofilm (solved by damped Newton to $|Q| < 10^{-10}$ every step), then the mechanics once at $T^* = 1$: small strain, linear elasticity, growth eigenstrain $(\alpha - 1)\mathbf{I}$, stiffness $(\phi^2 + f)E_{\mathrm{bio}}$, $\nu = 0.3$, substrate fixed. These are checks of the kinematics; the full model uses the mechanics of Klempt et al. 2024.

7 Verification

check	result	file
residual = gradient of the functional, row by row, against an independent hand-written residual	composition rows to round-off; phase row differs by the term the hand version approximated	`unified_two_level_variational_check.py`
one node, no coupling: composition rows equal the repository's point-model residual (`core_hamilton_1d.residual`)	to round-off, every row	same
uniform field, $\beta_s > 0$, no growth: every node equals the point model solved to convergence	8.9×10^{-11} (two-level), 1.6×10^{-13} (one-level)	`unified_two_level_1d.py`, `unified_1d_monolithic.py`
discrete energy, every step, closed system	decreases at every step; the only increases are at the antibiotic switch	`unified_two_level_1d.py` (two-level), `unified_two_level_fuel_1d.py` (with nutrient)
Newton	$\lvert R\rvert_\infty < 10^{-8}$ every step, at most 31 iterations	`unified_two_level_1d.py`
time step, 1D	60 against 120 steps: $\alpha - 1$ changes by 0.7 %	same
mesh, 3D lumped	16^3, 24^3, 32^3: edge displacement within 3 %	`fuel_front_3d.py`
mesh, 3D five species	16^3 against 24^3: within 8 % (constant c^*), 9 % (local nutrient)	`fuel_front_3d_species.py`
point model inside the 3D runs	residual $\leq 4.5 \times 10^{-13}$ every step	same

8 Results

All runs use the point model calibrated for the commensal flow condition (CH) of Heine et al., started from the calibration's Day-1 state put to the inner liquid $s_0 = 0.1$, with the calibration's time link (0.2375 units of the point model's time per T^*, $T^* = 1$ being Day 21). Antibiotic parameters are assumed: either the same susceptibility for all species ($b_i = 1$, $\alpha^* = 0.65$) or the anaerobes (*V. dispar*, *F. nucleatum*, *P. gingivalis*) ten times more susceptible ($\alpha^* = 0.32$); switched on at $T^* = 0.5$.

1D, two-level model with the growth reaction. Seed $\phi = 0.9$ on $x \leq 0.25$, outer liquid 10^{-4}, nutrient supplied at $x = 1$; $\lambda = 1$, $\beta = \beta_s = 0.02$, $L_0 = 0.7$, $Y = 1$, $c_{\text{stop}} = 0.1$, $D = 1$.

	$\alpha - 1$ at $x = 0$, 0.2	c at $x = 0$, 0.25	$\phi(0)$	living share at $x = 0$
no antibiotic	0.137, 0.118	0.938, 0.949	0.77	0.53
antibiotic from $T^* = 0.5$ (all species)	0.116, 0.102	0.951, 0.959	0.67	0.31

The antibiotic lowers the growth by 15 %, the living share from 0.53 to 0.31 and the phase itself, and leaves more nutrient. With a closed boundary the discrete total energy decreases at every step (largest change -8.6×10^{-3}). The edge in the current configuration (Proposition 2, $L_0 = 3$) moves from 0.25 to 0.80 by $T^* = 1$ with a rich supply ($c_{\text{stop}} = 0.1$), to 0.35 with a poor one ($c_{\text{stop}} = 0.5$), and to 0.62 with the antibiotic from $T^* = 0.5$.

1D, sensitivity. With the growth energy of Remark 2 in place of the reaction ($k_\alpha = 2$ as base), one parameter at a time, $T^* = 1$.

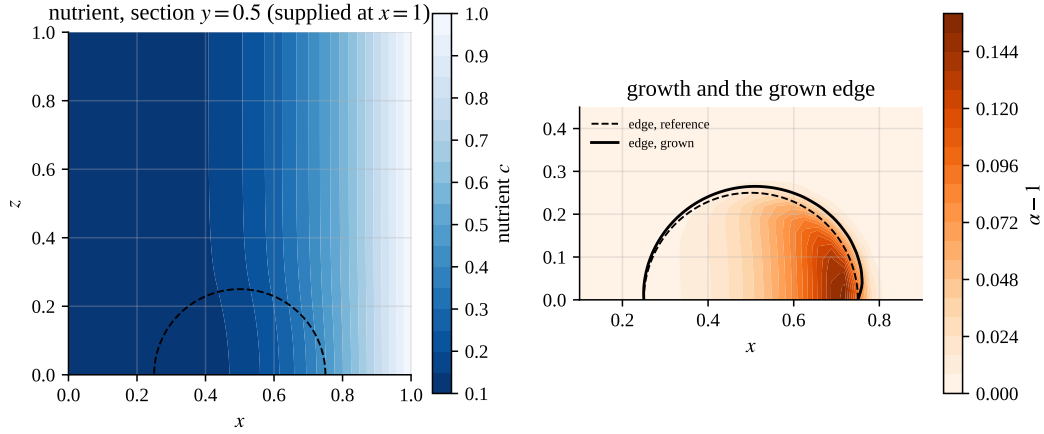
parameter	value	$\alpha - 1$ at $x = 0$	ϕ at $x = 0,\ 0.3,\ 1$	living, $x = 0$	antibiotic: $\Delta(\alpha - 1)$, $\Delta\phi(0)$
λ	0.3	0.236	$0.90,\ 0.31,\ 10^{-4}$	0.79	$-5\,\%$, -0.04
	1 (base)	0.183	$0.81,\ 0.28,\ 10^{-4}$	0.59	$-11\,\%$, -0.12
	3	0.125	$0.60,\ 0.21,\ 10^{-4}$	0.54	$-18\,\%$, -0.22
k_α	0.5	0.043	$0.77,\ 0.27,\ 10^{-4}$	0.52	$-16\,\%$, -0.11
	8	1.339	$0.99,\ 0.68,\ 10^{-4}$	0.94	$-1\,\%$, 0.00
β	0.005	0.183	$0.81,\ 0.17,\ 10^{-4}$	0.59	$-12\,\%$, -0.12
	0.08	0.171	$0.67,\ 0.34,\ 3\times10^{-4}$	0.59	$-10\,\%$, -0.10
β_s	0.005, 0.08	0.183	$0.81,\ 0.28,\ 10^{-4}$	0.59	$-11\,\%$, -0.12

The outer liquid stays empty in every case. λ sets how strongly the antibiotic acts on the growth and on the phase; β sets the spreading; β_s does not matter for a seed of uniform composition. The row $k_\alpha = 8$ shows the defect of the growth energy that led to the reaction form (Remark 2).

2D and 3D, direction of the front. Half disc (2D) or half ball (3D) of radius 0.25 on the substrate, nutrient supplied on the face $x = 1$ only, the liquid starting at c_{stop}; lumped composition (living share 0.5); $L_0 = 2$, $D = 0.2$.

	mesh	towards the nutrient	away	ratio
2D, 30°	48^2, $D = 0.2$	0.0180	0.0053	3.4
2D, 30°	48^2, $D = 1$	0.0532	0.0377	1.4
3D, elevation 30°	16^3	0.0190	0.0049	3.9
3D, elevation 30°	24^3	0.0206	0.0050	4.1
3D, elevation 30°	32^3	0.0213	0.0051	4.2

Outward displacement of the edge at $T^* = 1$. The edge advances towards the nutrient by the growth kinematics alone, the more the slower the supply.

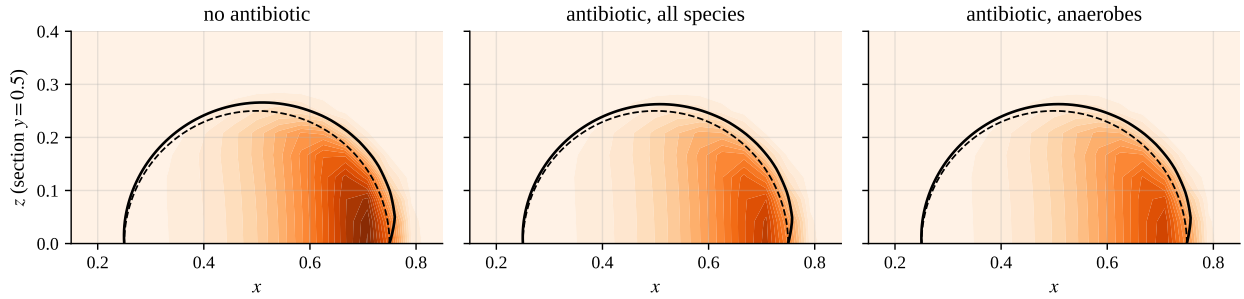


3D, section $y = 0.5$: nutrient (left) and growth with the grown edge (right), 24^3.

3D with the five-species point model. As above with the calibrated point model at every node of the biofilm, 24^3, $D = 0.2$, $L_0 = 2$.

	living share, centre	$\alpha - 1$, nutrient side / far side	edge 30°, nutrient side / far side
no antibiotic	0.518	0.116 / 0.005	0.0186 / 0.0047
antibiotic, all species	0.281	0.086 / 0.003	0.0137 / 0.0033
antibiotic, anaerobes	0.092	0.057 / 0.002	0.0090 / 0.0019
with the local nutrient in the point model (interaction $c^* c(\mathbf{x})/c_{\text{ref}}$, $c_{\text{ref}} = 1$):			
no antibiotic	0.507	0.135 / 0.005	0.0216 / 0.0053
antibiotic, all species	0.332	0.111 / 0.004	0.0176 / 0.0043
antibiotic, anaerobes	0.354	0.109 / 0.004	0.0174 / 0.0044

With the local nutrient the composition varies in space: on the nutrient side ($c = 0.67$ for $D = 1$) *V. dispar* rises to 0.19 and the living share falls to 0.37, on the far side ($c = 0.38$) they are 0.16 and 0.44; where the nutrient is scarce the interactions are weak and more cells stay alive. The antibiotic reduces the edge displacement by 26 % (all species) and 52 % (anaerobes) with the uniform interaction, by 19 % and 20 % with the local nutrient. Field and point model share one time here ($T^* = 1$ is Day 21 for both); the first 3D runs had the point model only at about Day 5 and gave much weaker antibiotic effects.



3D, five species, local nutrient, section $y = 0.5$: $\alpha - 1$ and the grown edge without antibiotic, with a broad antibiotic and with an anaerobe-specific one.

9 Finite element implementation

The model is written as an Abaqus user element (`p3/uel/unified_uel.f90`): an 8-node hexahedron with the displacement, ϕ and c at the nodes (Abaqus degrees of freedom 1–3, 11, 12) and the composition, the growth $\xi = \alpha - 1$ and a start flag at the $2 \times 2 \times 2$ Gauss points. The phase and nutrient equations and the growth (12) are solved implicitly, with the composition advanced once per increment by the point model of Klempt et al. 2026 at every Gauss point where ϕ has passed $\phi_{\min} = 0.05$ (staggered, the nutrient of the start of the increment). The mechanics is finite strain, $\mathbf{F} = \mathbf{F}_e \mathbf{F}_g$, $\mathbf{F}_g = \alpha \mathbf{I}$, with the energy per reference volume

$$W = \alpha^3 (\phi^2 + f) \left[\frac{\mu}{2} \left(J_e^{-2/3} \mathbf{I} : \mathbf{C}_e - 3 \right) + \frac{K}{2} (J_e - 1)^2 \right],$$

the mechanical term of Klempt et al. 2024 (Eq. 20) with a volumetric penalty in place of the incompressibility constraint; small strain is an option. The tangent is written in closed form. An antibiotic can be switched on at a given time through two element properties. The field and the point model share one time: $T^* = 1$ is Day 21 for both.

Abaqus is not available in the cloud; the element is checked by calling its entry point with Abaqus's argument list (`p3/uel/emulate_abaqus.py`). The run inside Abaqus is the next step.

check	result
one element, uniform field: Gauss-point composition against the point model solved to convergence	7.8×10^{-16}
1D column, 20 and 40 elements, against the 1D finite-difference model (Section 8)	$\alpha - 1$ 0.1371 / 0.1370 (1D: 0.137), $\phi(0)$ 0.776 / 0.772 (0.77), $c(0)$ 0.936 / 0.938 (0.938)
repeated calls in one increment (as Abaqus iterates)	residual, tangent, state identical
closed-form tangent against central differences	8.7×10^{-10} (all rows), 7.2×10^{-11} (mechanics, displacement gradients about 0.3)
free growth $\mathbf{u} = (\alpha - 1)\mathbf{X}$	mechanics residual 2×10^{-21}
3D seed, 16^3, small strain, against the finite-difference 3D model ($\lambda = 0$)	ϕ at the centre 0.778 / 0.779; edge 0.0135 / 0.0126; antibiotic $-23\,\%$ / $-24\,\%$
finite against small strain, 8^3	edge 0.0141 against 0.0143 ($\alpha - 1 \approx 0.07$)
element with $\lambda = 1$, $\Psi_{\text{ref}} = 0.32$, 16^3, against the finite-difference 3D model	volume 0.0346 $\to$ 0.0304 ($-12\,\%$), finite differences 0.0351 $\to$ 0.0309 ($-12\,\%$); von Mises mean $-39\,\%$
large growth ($\alpha - 1 = 0.28$), finite against small strain, 8^3	edge 0.0507 against 0.0560 (small strain 10 % larger), von Mises mean $3\,\%$
nutrient in the current configuration: uniform expansion, stretched bar	amount conserved, $c = 1/1.331$; flux DJ/F_{11}^2; 8^3 seed: edge $-4\,\%$, von Mises $-6\,\%$
$\nu = 0.49$, 3D seed, von Mises mean (8^3 / 16^3)	fully integrated 0.0232 / 0.0201, mean dilatation 0.0134 / 0.0163 ($\nu =$ 0.3: 0.0130 / 0.0144)

3D with the composition acting on the phase ($\lambda = 1$). Finite differences, 16^3, point model converged at every node and step. The absolute level of the point model's energy is not fixed by the calibration (only its derivatives enter the point model), and with $\lambda > 0$ it pushes the phase down: without a reference the biofilm shrinks by 16 % without antibiotic. With the reference $\Psi_{\text{ref}} = 0.32$ (the time average of the untreated point model's energy) the untreated biofilm stays near $\lambda = 0$, and the antibiotic shrinks the biofilm itself.

	biofilm volume (reference)	ϕ, centre	edge 30°
$\lambda = 0$, no antibiotic	0.0361	0.78	0.0126
$\lambda = 0$, antibiotic	0.0361	0.78	0.0096
$\lambda = 1$, $\Psi_{\text{ref}} = 0.32$, no antibiotic	0.0351	0.78	0.0115
$\lambda = 1$, $\Psi_{\text{ref}} = 0.32$, antibiotic	0.0309 ($-12\,\%$)	0.67	0.0083

Ψ_{ref} is a new parameter, the energy of forming the biofilm; the time average is one choice, and it should be fixed by the measured volume.

10 Growth limited by pressure

Added on 16–20 October 2026. The growth rate of the element can be scaled by a homeostatic-pressure gate, an assumption of this work and not taken from a paper:

$$\dot{\xi} = g\,\dot{\xi}_0, \qquad g = \min\bigl(1, \max(0,\, 1 - p/p_h)\bigr), \qquad p = -\tfrac{1}{3}\operatorname{tr}\boldsymbol{\sigma}, \tag{13}$$

with $\dot{\xi}_0$ the growth rate of Section 9 and p the Cauchy pressure, positive in compression. The pressure is taken from the start of the increment, so g is constant within an increment and the tangent keeps its form; since ξ never decreases and $g \geq 0$, the gate cannot oscillate and p overshoots p_h by at most one increment of growth. An implicit gate (pressure of the end of the increment) is an option. Without the gate every earlier result is unchanged to the last bit.

check (one element, Klempt et al. 2024 limit, $E = 10\,\mathrm{Pa}$, $\nu = 0.49$)	result
free growth, gate active	$\alpha - 1 = 1.100001 \times 10^{-3} = k_\alpha T$, equal to the run without the gate to 10^{-14}
all $\mathbf{u} = 0$, $p_h = 0.25\,\mathrm{Pa}$, against the ODE $\dot{\alpha} = k_\alpha g(p(\alpha))$	error 1.25, 0.62, 0.31, 0.15×10^{-5} at 11, 22, 44, 88 increments (first order)
homeostatic limit, $T = 4$	growth stops at $\alpha - 1 = 4.9993 \times 10^{-4}$ (exact 5.0000×10^{-4}), $p = 0.24997\,\mathrm{Pa}$
tangent with $g = 0.5$ against central differences	3.8×10^{-10}

Cervical line of a tooth-like collar. A 90° sector of a 0.25 mm layer on a collar whose radius grows from 2.05 to 3.05 mm over its 2 mm height (a crown; slope 38° at the base), inner face bonded, bottom $u_z = 0$, outer face free, nutrient from the outer face, seed 0.125 mm. Band: the seed elements with $z \le 0.25$ mm; core: the seed with $0.25 \le z \le 1.75$ mm; ratio = band mean von Mises over core mean. Without the gate the band is the most stressed region of the seed:

	without the gate			$p_h/E = 0.05$		
mesh	ratio	band vM$/E$	max p/E	ratio	band vM$/E$	max p/E
$2 \times 6 \times 8$	1.22	0.158	0.186	0.59	0.036	0.049
$4 \times 12 \times 16$	1.36	0.220	0.349	0.65	0.041	0.052
$6 \times 18 \times 24$	1.37	0.234	0.442	0.67	0.042	0.053

Without the gate the ratio and the band mean settle with the mesh, but the largest pressure keeps rising at the corner of the elastic problem. With the gate the ratio, the band mean and the largest pressure ($= p_h$) are converged to 3 % at the third level, because the growth that loads the corner is capped. Clamping the bottom or fixing the outer face changes the numbers but not the conclusion (`p3/uel/README.md`).

Which p_h. The literature does not fix p_h. Over $p_h/E = 0.02$ to 1 the ratio rises from 0.54 to 1.26 and passes 1 between 0.2 and 0.5. Let p_0 be the band mean pressure without the gate. In four cases that change p_0 by a factor of 3 (the case above, half the growth time, half the bulge, the bottom clamped) the ratio crosses 1 at $p_h/p_0 = 1.0$, 1.2, 1.0 and 1.4 (Fig. 1): the gate removes the cervical concentration when p_h is below about the pressure that growth alone builds there. p_h and E need not be known separately.

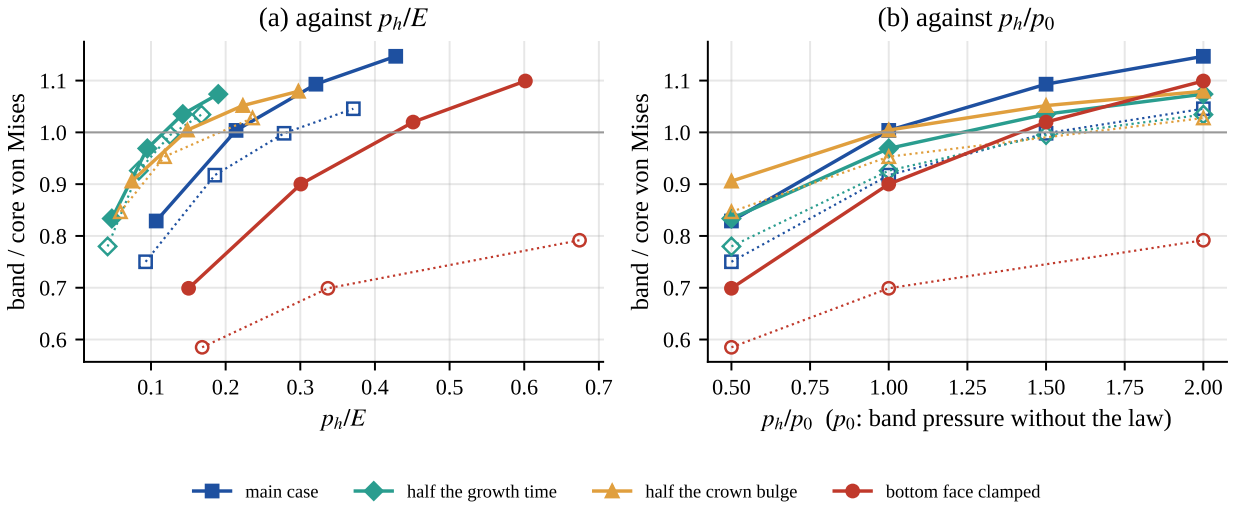


Figure 1: Ratio of band to core von Mises stress against p_h/E (left) and against p_h/p_0 (right) for four cases; $4 \times 12 \times 16$ filled, $2 \times 6 \times 8$ open.

Posterior spread. The interaction parameters of the point model are uncertain (thesis, Chapter 3). Particles of the final posteriors (200 per condition) were run through the same collar: 30 random particles each for

CH, DS and DH at $2 \times 6 \times 8$, 10 for CH at $4 \times 12 \times 16$, and the MAP. The coefficient of variation of the band von Mises stress is 19, 12 and 4 % (CH, DS, DH) without the gate and 1.5, 1.0 and 0.5 % with $p_h/E = 0.05$; at $4 \times 12 \times 16$ (CH) 10 % and 0.5 %. The ratio stays below 1 for every particle with the gate (0.61–0.68 at $4 \times 12 \times 16$) and above 1 without it (1.34–1.37). The MAP is an outlier without the gate (band vM$/E$ 0.50 against at most 0.26 for the particles). The gate absorbs the posterior spread of the composition, so the conclusion of this section does not hinge on the calibration.

Illustration without experimental data. The collar was also run with the four-species synthetic case of Fritsch et al. 2025 (their true parameters, $c^* = 25$; the fifth species absent), so that it can be shown without the calibration (Fig. 2; 3D videos in `assets/anim_cervical_f4_3d.mp4` and, for the cube, `assets/anim_law_3d.mp4`). Growth is large in this case (displacement up to 0.31 mm in the 0.25 mm layer). Without the gate the largest von Mises stress in the section is $0.68E$ at the bonded base; with $p_h/E = 0.05$ it is $0.12E$.

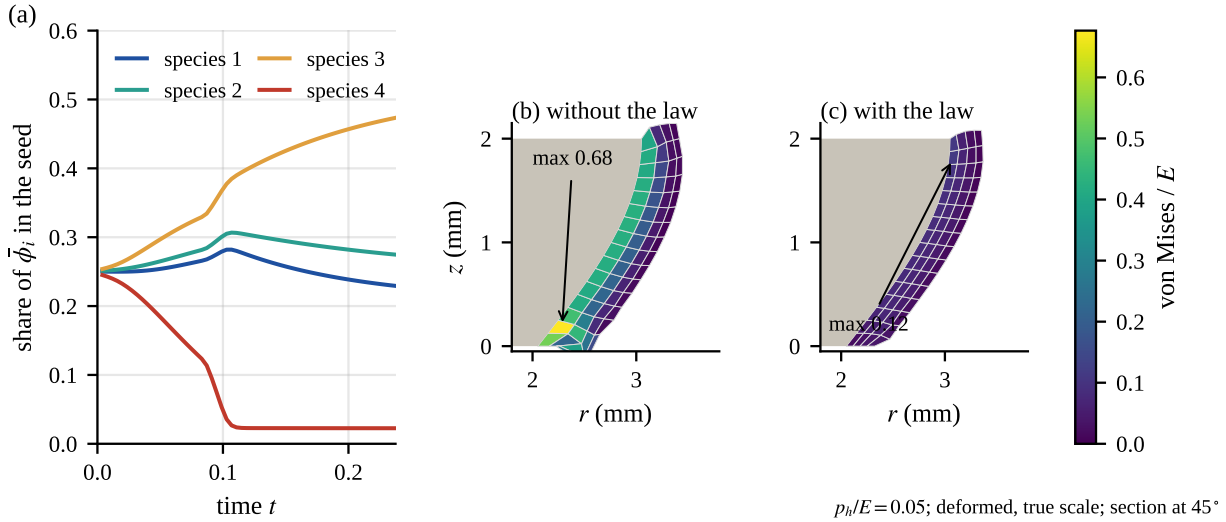


Figure 2: Synthetic four-species case: (a) shares of $\bar{\phi}_i$ in the seed over time; (b), (c) von Mises stress on the section at 45° at the end, deformed at true scale, without and with the gate ($p_h/E = 0.05$).

11 Assumptions and open points

assumption	reason	tested
two levels	one level fills the outer liquid (Section 1)	yes
weight $h(\phi) = \phi^2$	with $h = \phi$ the composition force does not vanish in the outer liquid and it fills again; ϕ^2 is the weight of the mechanical term of Klempt et al. 2024	yes
$\phi \leq 1$ by the one-sided barrier $K_\phi/(1-\phi)^2$	the phase equation has no upper bound (Klempt et al. 2024 neither); ϕ reached 2.7 at large growth without it	yes
growth as a reaction with the living biomass as catalyst	Remark 2	yes
ideal-solute energy of the nutrient, linear reaction kinetics	simplest forms consistent with the principle	sensitivity only
λ, β_s, L_0, Y, c_{stop}, D	not from a paper; λ sets how strongly an antibiotic acts on the phase, β_s does not matter for a seed	λ, β_s yes; others no
time link of the composition (0.2375 per T^*)	the calibration's; in the model it is the ratio η/η_ϕ and can be fixed by the measured volume	no
local nutrient in the point model, $c^* c/c_{\text{ref}}$ with $c_{\text{ref}} = 1$	the element's option; the relation of c^* to c should come from the nutrient level of the calibration	no
front term of Klempt et al. 2024 left out	not derivable from a potential; the edge moves by the kinematics (Proposition 2)	1D, 2D, 3D
mechanics: penalty in place of the incompressibility constraint, with the mean dilatation at $\nu = 0.49$; nutrient in the reference configuration by default	finite strain in the element (Section 9), the current configuration as an option; the finite-difference 3D runs are small strain	finite against small strain, $\nu = 0.3$ against 0.49, reference against current configuration
reference energy Ψ_{ref} for $\lambda > 0$	the calibration does not fix the level of the point model's energy	one choice

Open: β_s in 3D (volumetric locking at $\nu = 0.49$ is removed by the mean dilatation: in a bilayer strip the tip deflection is 1.07 times beam theory on a coarse mesh, against 0.67 fully integrated, and in the 3D seed the fully integrated element overstates the von Mises stress by 23 % even at 16^3); the run inside Abaqus; the limit $\phi \to 0$ of the composition equations (regularised by ε); existence and uniqueness (not attempted); a reader other than me; the overlap with F. Klempt's unpublished two-species work (only published sources were used; to be discussed before any publication). The checks establish that the equations are the ones the energy and dissipation imply, that the solver solves them, and that the results do not depend on the mesh or the time step at the level stated. They do not establish that the model is right: the only data used are the calibrated interactions of the point model. The measured volumes of Heine et al. and an antibiotic experiment would test the growth law and λ.